

INFOSYS VIRTUAL INTERNSHIP 6.0

FOOD TRENDS UNDERSTANDING CUSTOMER PREFERENCES

BATCH: 11 | GROUP: 1

TEAM: C

YEAR: 2026

Internship Duration: 8 Weeks

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1. Project Title

Infosys_FoodTrends: Understanding Customer Preferences

2. Project Objective

The primary objective of this project is to analyze restaurant data and derive insights that explain customer preferences and behavior. The project aims to understand how factors such as cuisine type, price range, online delivery, and customer engagement influence restaurant ratings and popularity. Another key objective is to present these insights in an interactive and visually appealing manner so that they can be easily interpreted by non-technical audiences, including media professionals and the general public.

Key objectives:

- Analyze customer preferences across different cuisines and regions
- Study customer engagement using ratings and vote counts
- Understand the impact of pricing on customer perception
- Communicate insights through interactive dashboards

3. Project Description

3.1 Overview

The Infosys Food Trends project was carried out as part of the Infosys Springboard Internship program and was executed as a team project involving five members. The dataset used for this project was sourced from Kaggle and contains detailed information about restaurants, including location, cuisine types, pricing, ratings, votes, and service availability. The project resulted in the development of seven interactive dashboards using Power BI, each designed to focus on a specific analytical dimension. Collectively, these dashboards provide a comprehensive understanding of food trends and customer preferences.

Overview highlights:

- Team-based project with five contributors
- Public dataset sourced from Kaggle
- Seven dashboards covering multiple analytical aspects
- Focus on real-world applicability

3.2 Project Approach

The project followed a well-defined and structured approach to ensure accuracy and quality. Initially, the dataset was collected from Kaggle and explored to understand its structure, attributes, and limitations. Data preprocessing was then carried out using Power Query to handle missing values, remove duplicates, and standardize categorical data. Additional calculated fields were created to support deeper analysis, such as price categories and engagement indicators. Once the data was cleaned and prepared, a data model was created in Power BI to establish proper relationships. The dashboard development phase involved selecting relevant KPIs, designing visuals, and adding slicers for interactivity. Throughout the project, dashboards were reviewed and refined through team discussions.

Step 1: Data Collection

The dataset used for this project was collected from Kaggle. It contained detailed information about restaurants, including location, cuisine types, pricing, ratings, vote counts, and service availability. The dataset was imported into Power BI for further processing and analysis.

Activities performed:

- Selected relevant dataset from Kaggle
- Imported dataset into Power BI
- Reviewed dataset structure and columns

Step 2: Data Cleaning and Preprocessing

The collected dataset contained missing values and inconsistencies, which were handled using Power Query. Duplicate records were removed, and categorical fields were standardized to ensure consistency. This step ensured that the dataset was clean and ready for analysis.

Activities performed:

- Handled missing and inconsistent data
- Removed duplicate records
- Standardized and formatted data fields

Step 3: Data Modeling

After preprocessing, the data was structured properly in Power BI to support efficient analysis. Calculated measures and KPIs such as average ratings, total votes, and restaurant counts were created using DAX.

Activities performed:

- Organized data for analysis
- Created calculated measures and KPIs
- Ensured accurate data relationships

Step 4: Dashboard Development

Interactive dashboards were developed using Power BI to visualize restaurant trends and customer behavior. Various charts and filters were used to present the data clearly and enable easy analysis.

Activities performed:

- Designed interactive dashboards
- Added charts, filters, and slicers
- Focused on key analytical areas

Step 5: Insight Generation and Validation

The final step involved analyzing the dashboards to identify trends and patterns. Insights were reviewed and validated through team discussions to ensure accuracy and clarity.

Activities performed:

- Analyzed trends and patterns
- Derived meaningful insights
- Validated dashboards and findings

3.3 Technology Used

The project utilized Power BI as the primary visualization tool due to its ability to handle large datasets and create interactive dashboards. Power Query was used for data cleaning and transformation, ensuring the dataset was analysis-ready. DAX was applied to create calculated measures such as average ratings and vote-based metrics. Microsoft Excel was used for initial inspection and validation of the dataset before integration into Power BI.

Tools and technologies:

- Power BI for visualization and analytics
- Power Query for data preprocessing
- DAX for calculated measures
- Microsoft Excel for data validation
- Power point to prepare ppt
- Word to prepare the report
- Microsoft Teams platform where classes were conducted
- GitHub & Gitlab to facilitate hosting, collaboration

3.4 Insights from the Dashboard

The analysis generated meaningful insights into customer behavior and restaurant performance. Restaurants are largely concentrated in metropolitan areas, indicating strong urban demand. North Indian and Chinese cuisines dominate customer preferences, while online delivery services significantly increase customer engagement. Higher-priced restaurants generally receive better ratings, suggesting a link between price and perceived quality. Additionally, restaurants with more votes tend to maintain more stable ratings compared to those with fewer reviews.

Major insights:

- Urban regions dominate restaurant presence
- Certain cuisines show consistently higher demand
- Online delivery boosts engagement
- Higher pricing correlates with better ratings
- Vote count impacts rating stability

Business Insights:

1. Urban areas provide better business expansion opportunities

The analysis shows that most restaurants are located in metropolitan cities, indicating higher customer demand and stronger business potential. Urban regions offer larger customer bases and higher purchasing power, making them ideal locations for restaurant growth and expansion.

2. Popular cuisines help attract more customers

Certain cuisines, such as North Indian and Chinese, show higher demand and availability. Restaurants that offer popular cuisines can attract more customers, improve sales, and remain competitive in the market by aligning their menu with customer preferences.

3. Online delivery improves customer reach and engagement

Restaurants that provide online delivery services have higher customer engagement and interaction. Offering customers, increase convenience, and improve overall revenue potential. delivery services allows businesses to reach more

4. Premium pricing strengthens brand image and customer perception

Higher-priced restaurants tend to receive better ratings, indicating that customers associate higher prices with better quality and service. Businesses can use appropriate pricing strategies to enhance brand value and customer trust.

5. Customer reviews improve business credibility and performance

Restaurants with higher vote counts have more reliable ratings, showing that customer feedback plays an important role in building trust. Encouraging customer reviews helps businesses improve reputation and attract more customers.

6. Data-driven analysis helps businesses make better decisions

The dashboards provide insights that help businesses improve pricing, service quality, and expansion strategies. Using data-driven insights enables restaurants to make informed decisions and improve overall performance.

3.5 Real-World Impact for Media and Public Communication

The findings of this project have strong relevance in real-world scenarios, particularly for media and public communication. Media organizations can use the insights to publish data-driven articles on food trends and customer preferences. Consumers can leverage the dashboards to make informed dining decisions. Restaurant owners and business stakeholders can use the analysis to improve service offerings, pricing strategies, and customer engagement. Overall, the project demonstrates how data visualization can effectively communicate complex information to diverse audiences.

Real-world impact:

- **Supports data-driven media reporting:**
The dashboards provide accurate and structured insights into restaurant trends, cuisine popularity, and customer behavior. Media organizations can use these insights to create data-driven reports, articles, and visual stories that reflect current food industry trends. This improves the credibility and reliability of media content by supporting it with factual data.
- **Enhances public awareness of food trends:**
The visualizations help present complex restaurant data in a simple and understandable format. This allows the public to easily understand trends such as popular cuisines, pricing patterns, and customer preferences. Increased awareness helps consumers make informed decisions when choosing restaurants and understanding market trends.
- **Aids business strategy and decision-making:**
Restaurant owners and business stakeholders can use the insights to improve their pricing strategies, menu offerings, and service quality. By understanding customer preferences and engagement patterns, businesses can make better decisions to improve performance, customer satisfaction, and profitability.
- **Demonstrates effective visual storytelling:**
The dashboards transform raw data into interactive visual representations that clearly communicate insights. This improves the ability to present analytical findings in a meaningful way, making it easier for both technical and non-technical audiences to understand key information and trends.

4. Timeline Overview

The project was completed over a total duration of forty days. The execution was divided into milestone-based phases, with each milestone allocated one week. This structured timeline ensured systematic progress, effective task distribution, and timely completion of deliverables.

Timeline highlights:

Week	Activities Planned	Activities Completed
1	Dataset selection, understanding project requirements, and defining objectives	Selected Kaggle dataset, analyzed data structure, and finalized project objectives
2	Data cleaning, preprocessing, and transformation	Cleaned dataset using Power BI Power Query, handled missing values, standardized columns
3	Data modeling and creation of calculated measures	Established data relationships and created DAX measures for key metrics
4	Design of core dashboards and KPI visuals	Developed KPI cards, bar charts, and slicers for overall performance analysis
5	Customer and restaurant performance analysis	Built dashboards for customer behavior and restaurant-wise performance
6	Food category and trend analysis	Analyzed popular cuisines, items, and ordering trends using time-based visuals
7	Insight generation and dashboard optimization	Derived key insights, improved dashboard layout, and optimized visuals
8	Final review and documentation	Validated results, prepared project report, and finalized dashboards

5. Project Execution

5a. Key Milestones

The first milestone focused on data collection and preprocessing to prepare the dataset for analysis. During the second milestone, the first three dashboards were developed, focusing on market presence, cuisine demand, and customer engagement. The third milestone involved completing five dashboards and refining previously developed dashboards. The final milestone was dedicated to completing all seven dashboards, validating insights, and optimizing performance.

Milestone Overview:

Milestones	Description	Date
Milestone 1	<u>Data Preparation & Transformation:</u> Collected dataset from Kaggle, cleaned and transformed data using Power BI Power Query, and created calculated columns and measures.	16-01-2026
Milestone 2	<u>Core Dashboard Development:</u> Designed interactive dashboards to analyze business performance, customer behavior, and restaurant performance using KPIs and trend visuals.	23-01-2026
Milestone 3	<u>Food & Category Trend Analysis:</u> Analyzed food categories, popular items, ordering patterns, and rating trends through category-wise visualizations.	30-01-2026
Milestone 4	<u>Strategic Insights & Optimization:</u> Derived actionable insights on restaurant consistency and customer satisfaction, optimized dashboards, and completed project documentation.	06-02-2026

Milestone summary:

Milestone 1: Data Preparation & Transformation

The first milestone focused on collecting and preparing the dataset for analysis. The dataset was sourced from Kaggle and contained restaurant, customer, order, and rating-related information. An initial exploration of the dataset was conducted to understand its structure, attributes, and data types.

Data cleaning was performed using **Power BI Power Query**, where missing values, inconsistent entries, and redundant records were identified and handled appropriately. Columns were standardized to maintain uniformity in formats such as categorical values, dates, and numerical fields. Unnecessary columns were removed to improve data model efficiency.

Additionally, calculated columns and DAX measures were created to support analytical requirements such as total revenue, order count, average ratings, and customer-related metrics. This milestone ensured that the dataset was accurate, consistent, and analysis-ready, forming a strong foundation for subsequent dashboard development.

Milestone 2: Core Dashboard Development

In the second milestone, the focus shifted to designing and developing the core dashboards in Power BI. A structured dashboard layout was planned to represent overall business performance, customer behavior, and restaurant performance in a clear and interactive manner.

Key visual elements such as **KPI cards, bar charts, line charts, and slicers** were implemented to present high-level metrics including revenue, total orders, customer count, and average ratings. Relationships between tables were verified to ensure accurate filtering and cross-visual interactions.

The dashboards were designed with usability in mind, enabling users to quickly interpret trends and compare performance across cities, restaurants, and time periods. This milestone resulted in functional dashboards that provided a comprehensive overview of the dataset.

Milestone 3: Food & Category Trend Analysis

The third milestone involved detailed analysis of food categories, popular items, and ordering patterns. Category-wise performance was examined to identify the most preferred cuisines and food items based on order frequency and customer ratings.

Trend analysis was conducted to study ordering behavior across different time periods, days, and customer segments. Visualizations were created to highlight variations in demand, peak ordering periods, and customer preferences. Rating trends were also analyzed to understand the relationship between food quality, consistency, and customer satisfaction.

This milestone helped uncover deeper insights into customer behavior and food trends, enabling a more granular understanding of what drives demand and performance in the restaurant ecosystem.

Milestone 4: Strategic Insights & Optimization

The final milestone focused on generating actionable insights and optimizing the dashboards for performance and clarity. Key insights related to restaurant consistency, quality stability, customer retention, and revenue concentration were derived from the analytical findings.

Dashboards were refined by optimizing visuals, reducing clutter, and improving interactivity through effective use of slicers and filters. Measures and calculations were validated to ensure accuracy and reliability of insights.

Finally, project documentation was prepared, summarizing the methodology, dashboards, challenges, and outcomes. This milestone ensured that the project delivered meaningful business insights and a polished analytical solution ready for evaluation and future enhancement.

5b. Project Execution Details

Step 1: Data Collection & Finalization

The dataset used in this project was sourced from Kaggle, which provided a comprehensive global restaurant dataset containing information such as restaurant names, locations, cuisines, ratings, votes, pricing levels, and service availability. After importing the dataset, the team collectively reviewed its structure, columns, and data quality. This step helped identify relevant attributes required for analysis and eliminate irrelevant or redundant fields.

The team also defined the project objectives and key business questions, such as identifying popular cuisines, understanding pricing impact on ratings, and evaluating customer engagement patterns.

Key activities performed:

- Dataset sourced and imported from Kaggle
- Dataset structure and column review
- Identification of relevant attributes for analysis
- Definition of project objectives and analytical goals

Step 2: Data Cleaning & Preparation

Data preprocessing was performed using the Power Query Editor in Power BI to ensure that the dataset was clean, consistent, and analysis-ready. Several transformation and cleaning operations were applied to improve data quality and usability.

Preprocessing steps performed in Power Query included:

- **Removal of unnecessary columns:**
Columns that were not relevant to the project objectives were removed to simplify the dataset and improve performance.

- **Handling missing and null values:**
Missing values in important fields such as ratings and votes were identified and handled appropriately to prevent errors during analysis.
- **Data type standardization:**
Columns were converted into appropriate data types such as text, numeric, and decimal to ensure accurate calculations.
- **Renaming columns for clarity:**
Column names were modified to improve readability and consistency.
- **Removing duplicate records:**
Duplicate entries were identified and removed to maintain data accuracy.
- **Filtering invalid or inconsistent data:**
Records with incorrect or incomplete values were filtered out to ensure reliable analysis.

Step 3: Data Modification and Feature Engineering

After preprocessing, additional modifications and feature engineering were performed to enhance the analytical capabilities of the dataset. New columns were created to categorize restaurants based on engagement, rating stability, and business priority.

These newly created columns helped improve analysis and supported predictive and strategic decision-making.

New columns created included:

- **Vote Bucket (Customer Engagement Category):** This column categorizes restaurants based on the number of votes received, helping identify low-engagement, medium-engagement, and high-engagement restaurants. This helps businesses understand customer interaction levels.
- **Quality Category (Rating Reliability Indicator):** This column categorizes restaurants based on rating stability and vote counts. Restaurants with higher votes and consistent ratings were classified as stable, while others were categorized as unstable or emerging.

- **Action Category (Business Priority Classification):** This column classifies restaurants into categories such as Promote, Monitor, Improve, or Maintain based on rating and engagement levels. This helps businesses identify which restaurants need attention or promotion.

These columns were created using DAX formulas and logical conditions to support strategic business analysis.

Step 4: Data Modeling & DAX Implementation

Measures Created using DAX

Measures were created to calculate key performance indicators (KPIs) used in dashboard visualizations. These measures allowed dynamic calculations based on filters and slicers.

- **Average Rating Calculation:** This measure calculates the average rating of restaurants and helps evaluate overall customer satisfaction.
- **Total Vote Count Calculation:** This measure calculates the total number of customer votes, which helps assess customer engagement and restaurant popularity.
- **Restaurant Count Aggregation:** This measure counts the total number of restaurants based on selected filters such as city, cuisine, or price range. This helps analyze restaurant distribution.
- **Engagement and Performance Metrics:** Additional measures were created to evaluate restaurant performance based on engagement, rating, and service availability.

Step 5: Dashboard Development and Visualization

After completing preprocessing and feature engineering, interactive dashboards were developed using Power BI. The dashboards were created milestone-wise, with each milestone focusing on specific analytical themes. The dashboards included visualizations such as bar charts, pie charts, maps, and line charts to present data clearly. Filters and slicers were added to allow users to explore data dynamically.

Dashboards developed included:

- Market and cuisine demand analysis
- Pricing and service impact analysis
- Customer engagement and rating behavior analysis
- Restaurant quality stability analysis
- Strategic action and business prioritization analysis

The dashboards were designed to ensure clarity, usability, and effective communication of insights.

Dashboard 1: Market Presence & Coverage Analysis

This dashboard provides insights into the geographical distribution of restaurants across different countries and cities. The analysis shows that restaurant presence is highly concentrated in specific countries and major metropolitan cities, indicating higher customer demand and business activity in urban areas. Certain cities act as major food hubs with a significantly higher number of restaurants compared to others.

Key Insights:

- Metropolitan cities have the highest restaurant concentration.
- Urban regions provide stronger business opportunities due to higher demand.
- Restaurant distribution is uneven, indicating potential expansion opportunities in underrepresented areas.
- Businesses can use this insight to identify high-growth and low-competition markets.

Dashboard 2: Cuisine Demand & Popularity Analysis

This dashboard analyzes the distribution and popularity of different cuisines across restaurants. The analysis shows that certain cuisines, particularly North Indian and Chinese, dominate in terms of availability and customer engagement. This indicates strong and consistent customer preference for these cuisine types.

Key Insights:

- North Indian and Chinese cuisines have the highest demand.
- Popular cuisines attract more restaurants and customers.
- Restaurants offering high-demand cuisines have better engagement potential.
- Businesses can optimize menu offerings based on customer preferences.

Dashboard 3: Customer Engagement & Rating Behavior

This dashboard focuses on customer engagement using metrics such as ratings and vote counts. The analysis shows that restaurants with higher vote counts tend to have more reliable and stable ratings. Restaurants with low votes often have less reliable ratings due to limited customer feedback.

Key Insights:

- Higher vote counts indicate higher customer engagement.
- Restaurants with more votes have more reliable ratings.
- Customer feedback plays a critical role in business credibility.
- Encouraging customer reviews improves restaurant visibility and trust

Dashboard 4: Pricing Strategy & Customer Perception

This dashboard analyzes the relationship between restaurant pricing and customer ratings. The analysis shows that restaurants in higher price ranges generally receive better ratings, suggesting that customers associate higher prices with better quality and service.

Key Insights:

- Premium-priced restaurants tend to receive higher ratings.
- Customers associate price with quality and experience.
- Pricing strategy influences customer perception.
- Businesses can use pricing to position their brand in the market.

Dashboard 5: Service Availability & Convenience Impact

This dashboard analyzes the impact of service features such as online delivery and table booking on customer engagement. The analysis shows that restaurants offering online delivery services receive higher customer engagement compared to those without delivery services.

Key Insights:

- Online delivery increases customer engagement and accessibility.
- Restaurants offering delivery services have competitive advantage.
- Convenience features improve customer satisfaction.
- Digital service adoption supports business growth.

Dashboard 6: Restaurant Consistency & Quality Stability Analysis

This dashboard evaluates rating stability and restaurant performance consistency. The analysis shows that restaurants with higher vote counts tend to have more stable ratings, indicating consistent quality and customer satisfaction.

Key Insights:

- High-engagement restaurants show stable performance.
- Low-engagement restaurants show rating variability.
- Consistent quality improves customer trust and loyalty.
- Businesses can monitor performance stability using engagement metrics.

Dashboard 7: Strategic Action & Restaurant Prioritization

This dashboard categorizes restaurants based on performance, engagement, and rating reliability to support business decision-making. Restaurants are classified into categories such as Promote, Improve, Monitor, and Maintain.

Key Insights:

- High-performing restaurants can be prioritized for promotion.
- Low-performing restaurants require improvement strategies.
- Medium-performing restaurants should be monitored.
- Businesses can allocate resources efficiently using performance classification.

Step 6: Documentation & Repository Management

The final step of the project involved organizing and documenting all project deliverables to ensure proper presentation and accessibility. All seven dashboards were integrated into a single Power BI file and reviewed to ensure consistency in design, layout, and functionality. This integration allowed the dashboards to provide a complete and structured analytical solution.

The project report was prepared to document the entire workflow, including data collection, preprocessing, feature engineering, dashboard development, and key insights. Presentation slides were also created to clearly explain the project objectives, methodology, and business insights.

To maintain proper version control and accessibility, the project files and documentation were organized and uploaded to a GitHub repository. This ensured secure storage, easy sharing, and future reference.

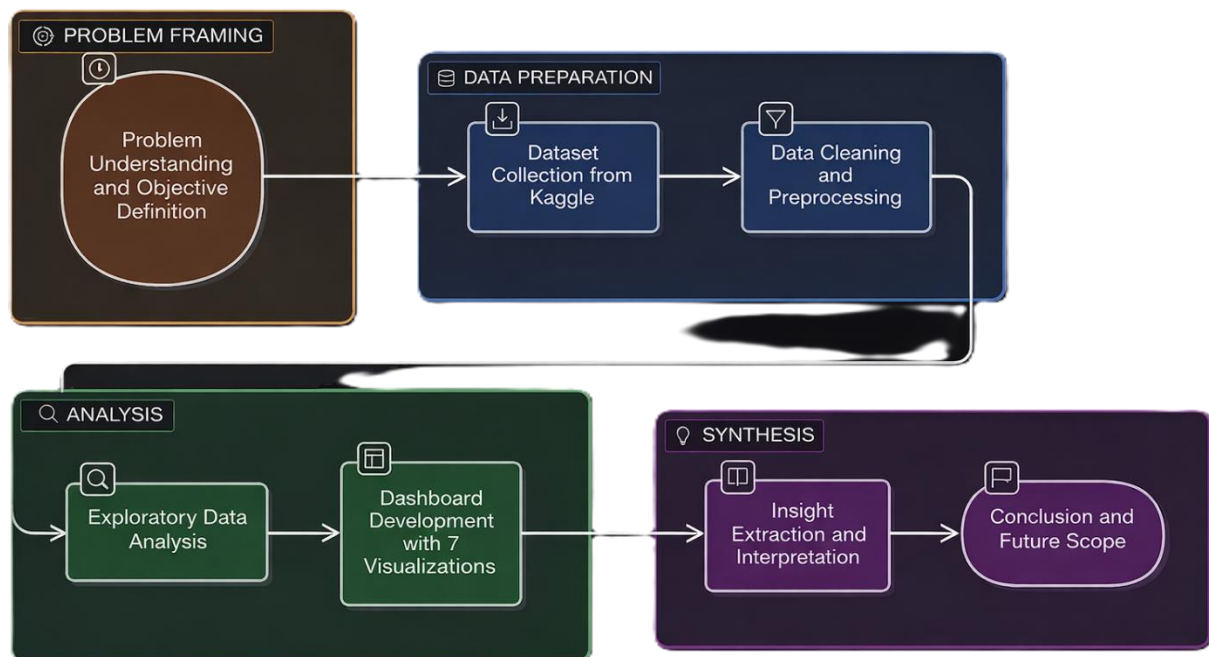
Key activities performed:

- Integration and validation of all dashboards in Power BI
- Preparation of final project report and presentation
- Organization of project files and documentation
- Uploading files to GitHub repository for version control and sharing

Execution Highlights:

- Strong team-based collaboration across all project stages
- Equal distribution and shared ownership of responsibilities
- Continuous review and iterative improvement process
- Consistent dashboard design and accurate analytical validation

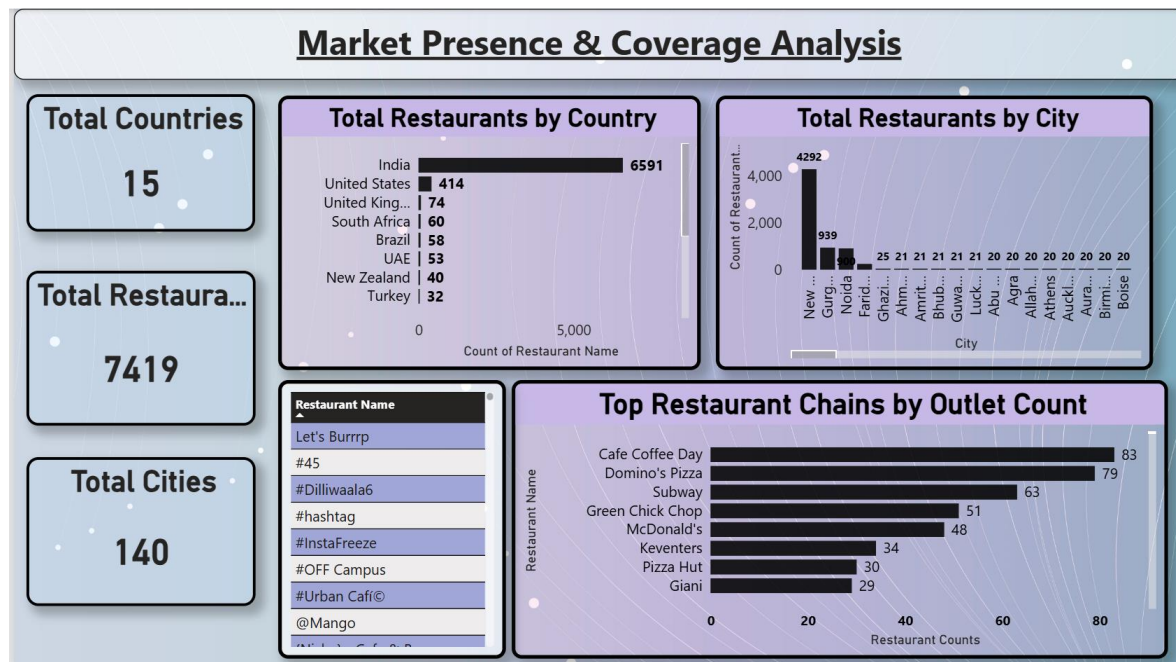
Work Flow



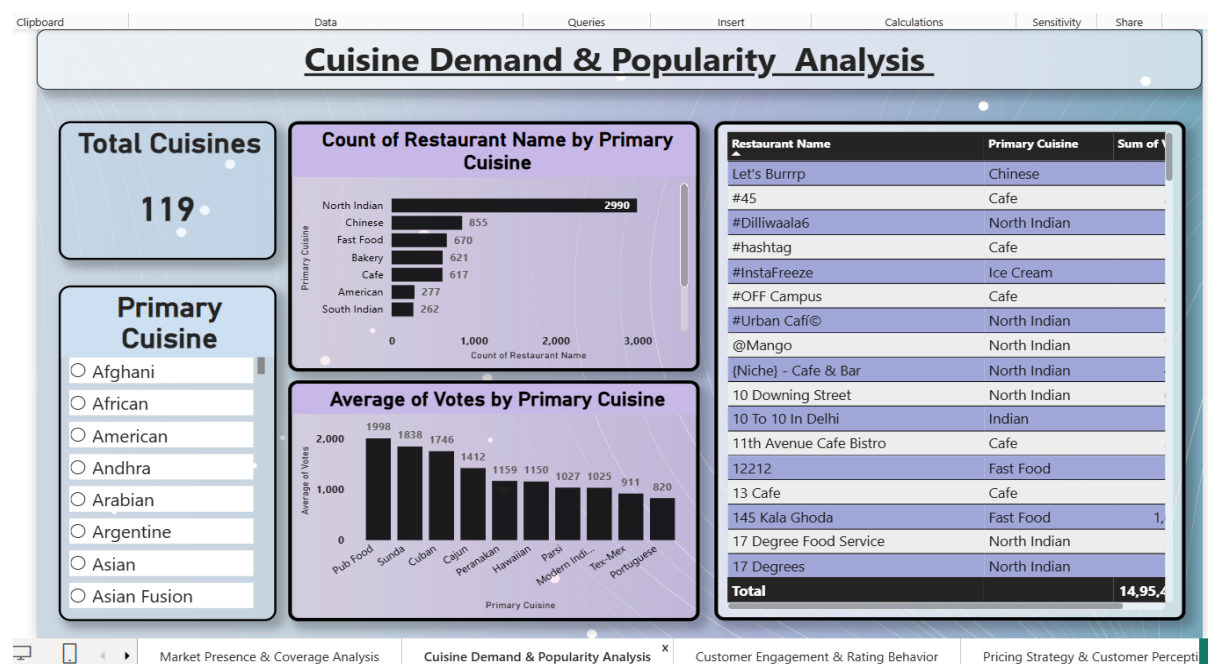
6. Snapshots / Screenshots

This section contains screenshots of all seven dashboards developed during the project. The screenshots visually demonstrate the KPIs, filters, and insights derived from the analysis and support the findings discussed in this report. Included dashboards are:

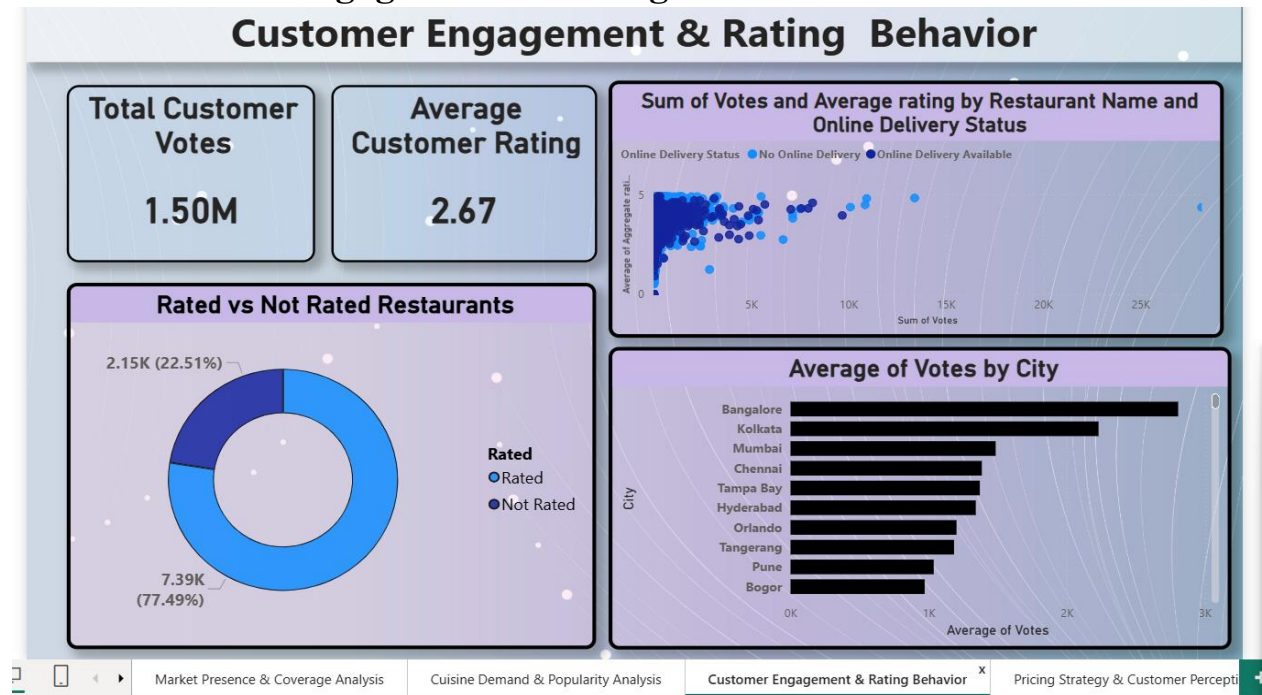
6.1 Market Presence & Coverage



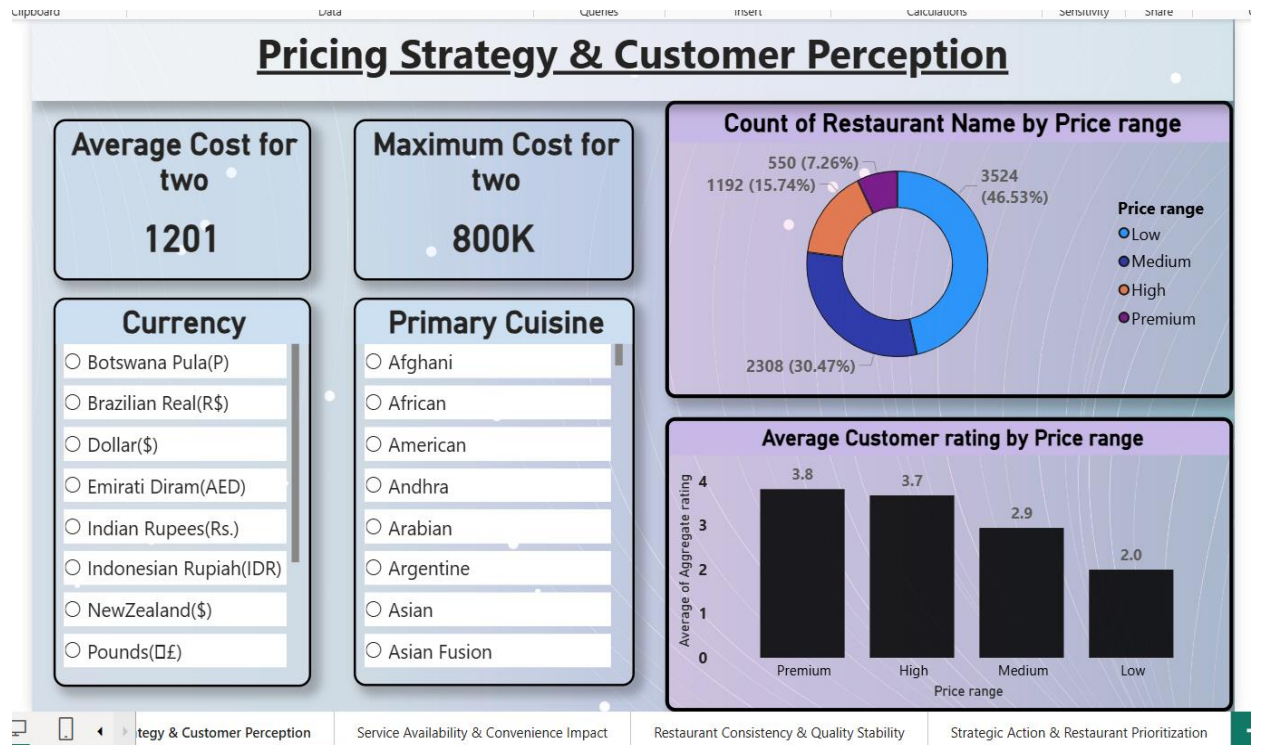
6.2 Cuisine Demand & Popularity



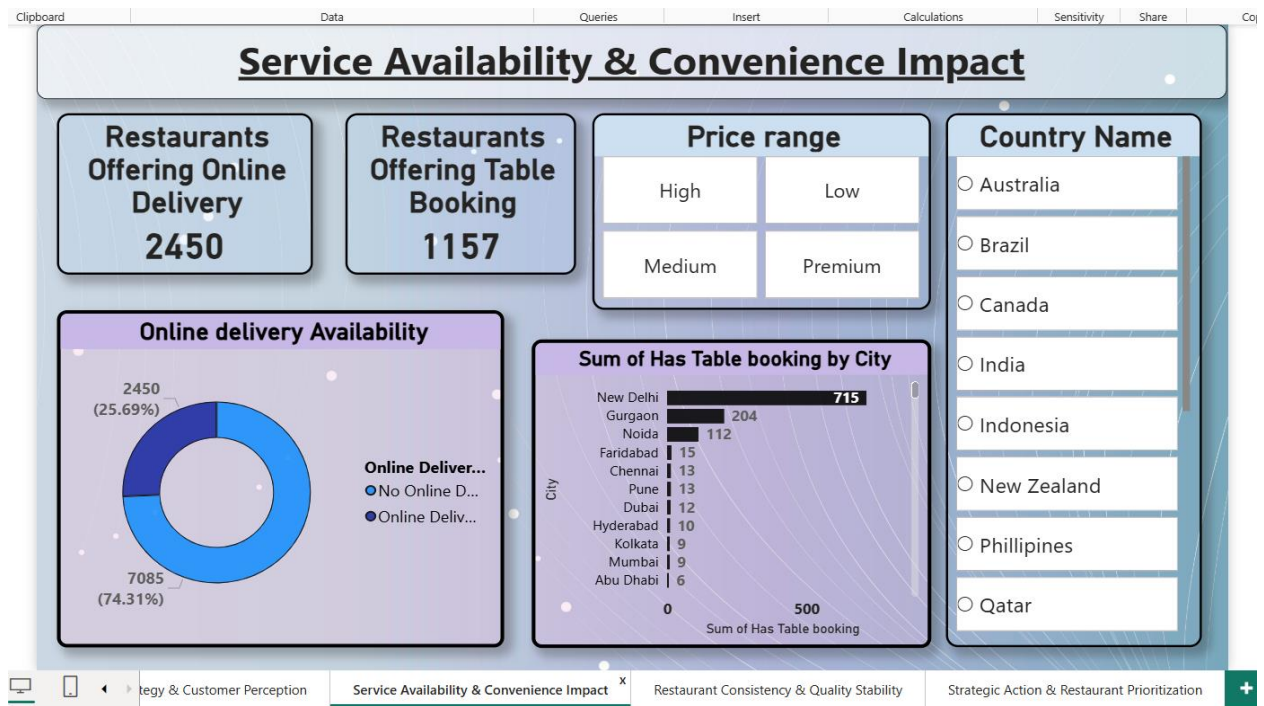
6.3 Customer Engagement & Ratings



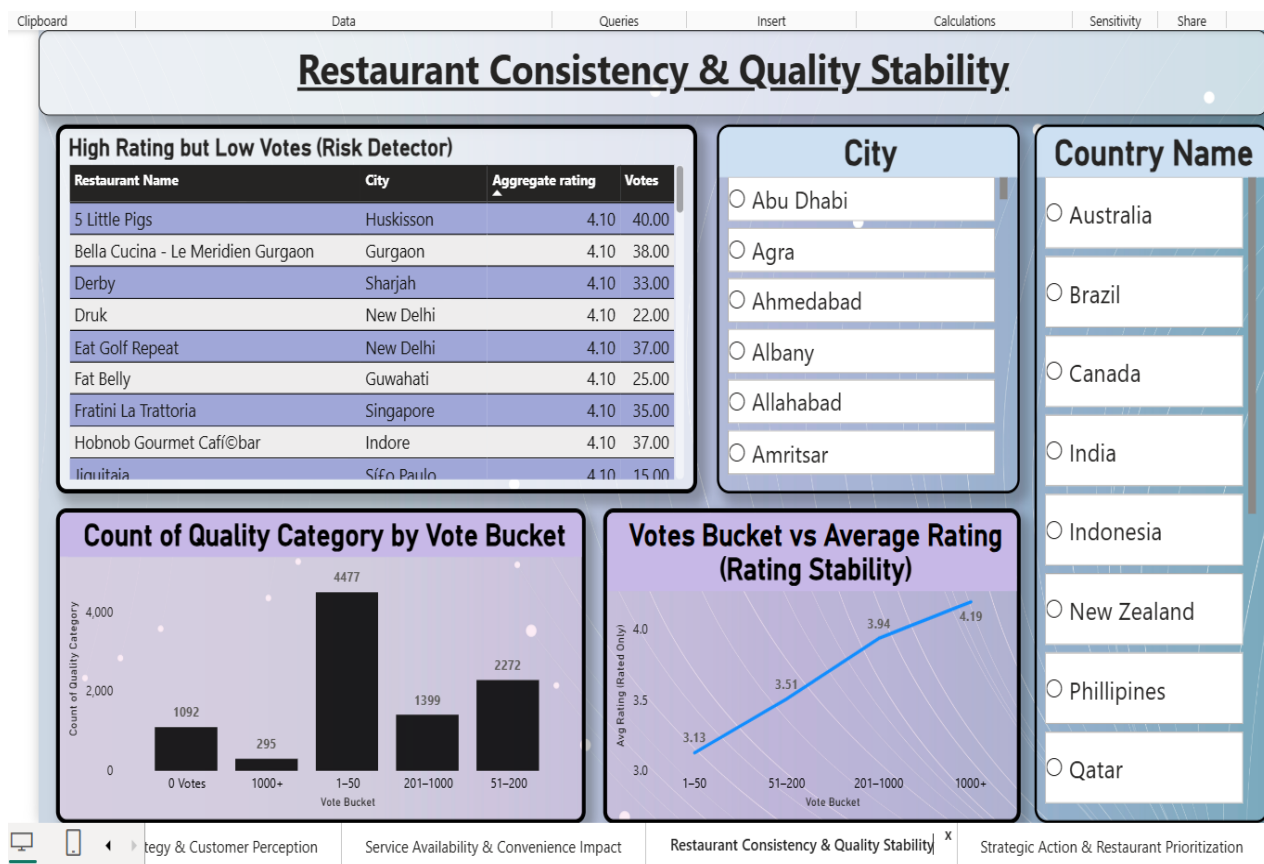
6.4 Pricing Strategy & Perception



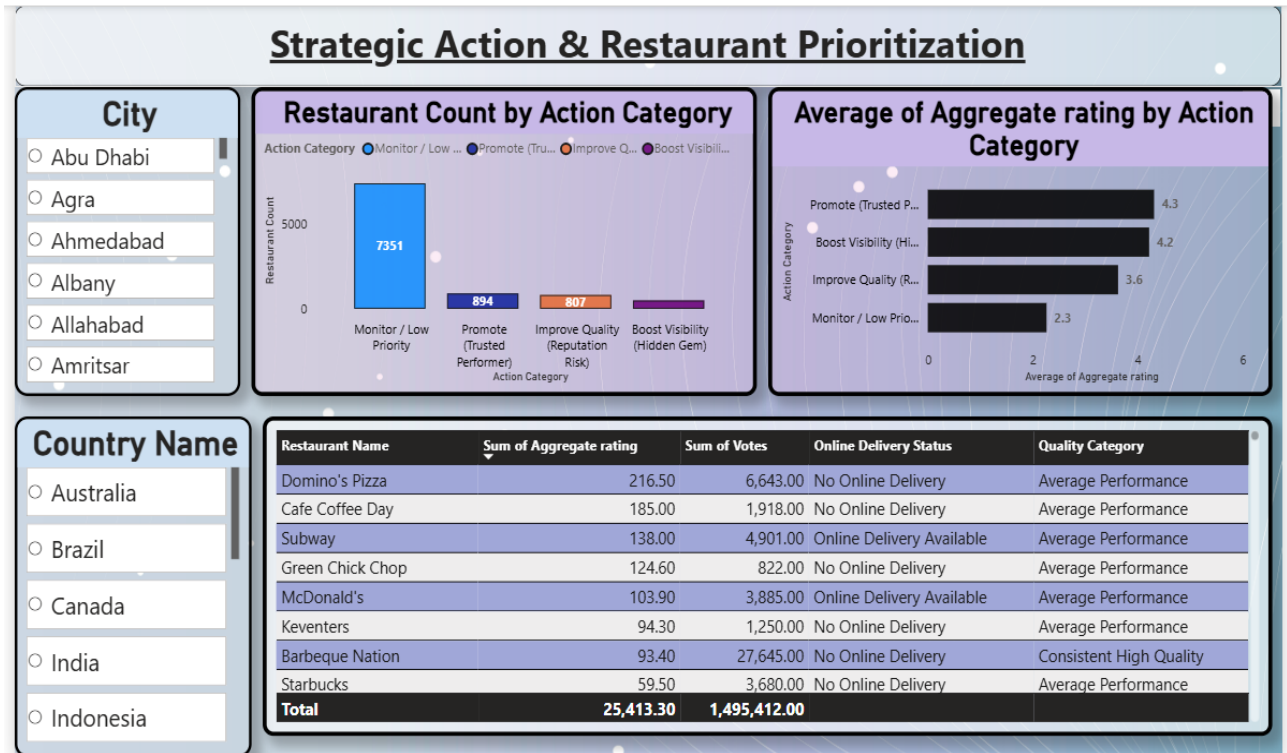
6.5 Service Availability Impact



6.6 Quality Stability Analysis



6.7 Strategic Restaurant Categorization



7. Challenges Faced

During the development of the project, several technical and analytical challenges were encountered. These included handling missing and inconsistent data entries, managing relatively large datasets within Power BI, selecting meaningful performance indicators, and interpreting rating trends where vote counts were low. Each of these challenges required careful analysis and structured problem-solving to ensure the reliability and clarity of the final dashboards.

Key Challenges and Resolutions:

- **Challenge 1: Inconsistent and Missing Data**

Resolution: Applied systematic data cleaning and preprocessing techniques using Power Query. Missing values were handled appropriately, inconsistent entries were standardized, and unnecessary columns were removed to improve data quality and consistency.

- **Challenge 2: Identifying Meaningful KPIs**

Resolution: Carefully analyzed business requirements to determine relevant metrics such as total revenue, order frequency, average ratings, and consistency scores. DAX measures were created to accurately calculate and represent these KPIs.

- **Challenge 3: Dashboard Clutter and Readability Issues**

Resolution: Improved dashboard layout by reducing redundant visuals, aligning elements properly, and incorporating slicers for dynamic filtering. This enhanced user experience and ensured better interpretability of insights.

- **Challenge 4: Performance Optimization in Power BI**

Resolution: Optimized DAX calculations, minimized redundant columns, and improved data model relationships to enhance dashboard performance and reduce loading time.

8. Learnings & Skills Acquired

8.1 Technical Skills

This project enhanced practical knowledge of Power BI, including dashboard creation, data visualization, and performance optimization. Hands-on experience was gained in Power Query for data cleaning and transformation, as well as in developing DAX measures and calculated columns. It also improved understanding of data modeling and building interactive dashboards.

8.2 Analytical and Problem-Solving Skills

The project strengthened analytical thinking by identifying meaningful KPIs and interpreting business trends from raw data. It improved the ability to analyze patterns, validate insights, and solve data-related challenges effectively.

8.3 Soft Skills and Team Collaboration

Working on this project improved communication, presentation, and time management skills. It also encouraged structured planning and collaborative discussion for refining dashboards and insights.

8.4 Domain Knowledge and Application

The project provided practical exposure to food industry analytics, including customer behavior analysis, revenue trends, cuisine preferences, and the importance of quality consistency in restaurant performance.

9. Testimonials from Team

Piyush Nitnaware:

Contributed to dataset collection, initial data understanding, and data cleaning using Power Query. Assisted in identifying missing values and standardizing columns for accurate analysis.

Vaishnavi Shah:

Worked on data transformation and creation of calculated columns and DAX measures. Contributed to defining key performance indicators such as revenue, order count, and average ratings.

Sam Joseph:

Led the development of core dashboards, including KPI visuals, bar charts, and trend analysis charts. Ensured proper data modeling within Power BI.

Raghav Vyas:

Focused on food category analysis, customer behavior insights, and restaurant performance comparison. Helped interpret rating trends and identify meaningful business insights.

Harini M:

Handled dashboard optimization, layout refinement, and report documentation. Contributed to insight validation, performance improvement, and final presentation preparation.

Team Reflection:

All team members expressed that the project provided valuable hands-on experience in data analytics and strengthened collaborative learning. The equal distribution of responsibilities, continuous peer discussions, and iterative review process were highlighted as key strengths that contributed to the successful completion of the project.

10. Conclusion

The *Food Trend Analysis and Restaurant Performance Dashboard* project demonstrates the effective application of data analytics and business intelligence techniques using Power BI. The primary objective was to transform raw restaurant and customer data into meaningful, actionable insights to support data-driven decision-making. This objective was successfully achieved through systematic data preparation, modeling, and interactive visualization.

The project involved thorough data cleaning and preprocessing using Power Query to ensure data accuracy and consistency. Calculated columns and DAX measures were developed to derive key metrics such as total revenue, order volume, average ratings, and customer demographics, enabling reliable and structured analysis.

The dashboards revealed important business insights, including revenue concentration in major cities such as Delhi (₹105M+), peak ordering on Fridays (34K+ orders), and an average customer age of 24 years. Cuisine analysis highlighted strong demand for North Indian and Chinese categories, while rating trends emphasized the importance of quality consistency for customer retention. Growth patterns indicated early expansion followed by market stabilization.

Overall, the project enhanced technical skills in data transformation, DAX, and dashboard design while strengthening analytical thinking and business interpretation abilities. It effectively showcases how Power BI can convert complex datasets into interactive dashboards that support strategic planning and performance evaluation.

11.Future Enhancements

1. Integration of Real-Time Data

The current dashboard uses a static dataset. In the future, integrating real-time data from restaurant platforms or APIs can provide continuously updated insights. This will help businesses monitor performance, track trends, and make timely decisions based on current data.

2. Implementation of Predictive Analytics

Machine learning models can be integrated to predict restaurant ratings, customer engagement, and demand trends. This will help businesses anticipate future performance and take proactive actions to improve service quality and customer satisfaction.

3. Expansion of Dataset with Additional Features

Future enhancements can include additional data such as customer reviews, revenue information, and time-based performance trends. This will provide deeper insights and improve the overall analytical capability of the dashboard.

4. Advanced Dashboard Interactivity and Visualization

Adding advanced features such as drill-down analysis, dynamic filters, and personalized views will improve user interaction. This will allow users to explore data more effectively and gain deeper insights.

5. Deployment on Cloud for Remote Access

The dashboard can be deployed using Power BI Service or cloud platforms to enable access from anywhere. This will help stakeholders monitor restaurant performance remotely and improve decision-making efficiency.

12. Acknowledgements

We would like to express our heartfelt gratitude to:

Infosys Springboard Team – for organizing this valuable internship opportunity and providing a structured learning platform.

Our Mentor – for her continuous guidance, constructive feedback, and encouragement throughout the project.

My Teammates – for their collaboration, support, idea sharing, and teamwork during all phases of the project.

We also sincerely appreciate the opportunity to work on a real-world dataset that allowed us to apply analytical concepts in a practical environment. The exposure to Power BI, data modeling, and structured project execution has significantly enhanced our technical knowledge and problem-solving skills.

The collaborative environment, regular reviews, and milestone-based execution helped us develop not only technical expertise but also essential skills such as communication, accountability, time management, and professional coordination.

This internship has been a significant step in our professional growth and technical journey.